

Hardware-Accelerated Orthogonal Polynomial Tensor Operators, Zero-Backpropagation Closed-Form Algebraic Solvers, and In-Situ Weight Surgery for Deep Neural Networks

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Abstract—Feed-forward network (FFN) layers in modern Transformer architectures allocate over 60% of total model parameters to high-rank linear matrix multiplications and activation functions. In resource-constrained embedded systems and edge accelerators, these massive parameter footprints and standard backpropagation requirements induce severe memory-wall bottlenecks. In this letter, we present an algebraic and hardware-accelerated paradigm that replaces standard multi-layer perceptrons with orthogonal Chebyshev polynomial tensor operators. We introduce: (1) an idempotent spectral projection manifold ($\Pi^2 = \Pi$) enabling zero-backpropagation closed-form algebraic solvers via regularized normal equations, running directly on edge hardware with sub-400 MB peak VRAM; (2) in-situ weight surgery on pre-trained large language models, evaluated across both compact embedded architectures (SmolLM2-135M with 56.2 ms convergence and 50% FFN reduction) and full 32-layer frontier reasoning systems (DeepSeek-R1-Distill-Llama-8B with 50.74 s streaming surgery, 3.49 billion parameter removal (-61.9% FFN), and 2.49× inference acceleration); (3) PolyFormer, a native ground-up architecture operating entirely on Chebyshev attention and tensor contractions; and (4) an asymptotic convergence study showing that low-parameter polynomial architectures fully surpass double-sized standard baselines under modest compute scaling. Complete open-source weights, PyPI packages, and Hugging Face checkpoints accompany this release.

Index Terms—Embedded systems, edge AI, in-situ weight surgery, orthogonal polynomials, zero-backpropagation, memory wall, Transformer accelerators.

I. INTRODUCTION

Modern Large Language Models (LLMs) dedicate the majority of their parameter footprint to feed-forward networks (FFN) [6], [7], [8]. Conventional fine-tuning and parameter adaptation rely exclusively on iterative gradient

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descent (AdamW), consuming substantial GPU memory and compute time. Recently, Kolmogorov-Arnold Networks (KAN) [11] and polynomial variants (e.g., ChebyKAN) demonstrated that parameterizing network edges with non-linear activation functions offers high theoretical expressivity. However, existing KAN formulations suffer from three fundamental systemic bottlenecks: (i) strict reliance on iterative backpropagation (AdamW) across millions of optimization steps, (ii) severe training latency overheads ($\sim 5\text{--}10\times$ higher than standard MLPs) and memory bloat from intermediate activation graphs, and (iii) the inability to compress or adapt pre-trained foundation models post-hoc without exhaustive re-training from scratch.

Building upon early mathematical foundations of idempotent permutations and algebraic invariant structures [2], [3], [4], we formulate feed-forward transformation as projection onto orthogonal Chebyshev polynomial manifolds [5]. In contrast to KANs, we prove that the optimal polynomial tensor coefficients can be solved analytically in closed form with **zero backpropagation** in a single $\mathcal{O}(1)$ step via regularized normal equations, enabling live in-situ model surgery directly on resource-constrained embedded accelerators and ground-up architectural synthesis.

II. ORTHOGONAL POLYNOMIAL TENSOR FORMULATION

Definition 1 (Orthogonal Chebyshev Tensor Operator). Let $X \in \mathbb{R}^{B \times d_{in}}$ be a normalized input tensor with $\tilde{x} = \tanh(x) \in [-1, 1]$. The K -th degree Chebyshev polynomial operator contracts input representations with a learned 3D coefficient tensor $\mathcal{C} \in \mathbb{R}^{(K+1) \times d_{in} \times d_{out}}$:

$$\mathcal{P}(X) = \sum_{k=0}^K \mathcal{C}_k \cdot T_k(\tilde{x}) + b \quad (1)$$

where $T_0(x) = 1$, $T_1(x) = x$, and $T_{k+1}(x) = 2xT_k(x) - T_{k-1}(x)$.

Theorem 1 (Zero-Backpropagation Closed-Form Solver). Given activation calibration pairs (X, Y) sampled from an arbitrary pre-trained network layer, the optimal polynomial coefficient tensor $\mathcal{C}^*$ minimizing the regularized Frobenius objective $\mathcal{L}(\mathcal{C}) = \|\Phi(X)\mathcal{C} - Y\|_F^2 + \lambda\|\mathcal{C}\|_F^2$ is uniquely determined by the normal equations:

$$\mathcal{C}^* = (\Phi(X)^T \Phi(X) + \lambda I)^{-1} \Phi(X)^T Y \quad (2)$$

where $\Phi(X) = [T_0(\tilde{X}), T_1(\tilde{X}), \dots, T_K(\tilde{X})]$. Furthermore, projecting onto the rank- r SVD subspace $\Pi = V_r V_r^T$ guarantees exact idempotence:

$$\Pi^2 = (V_r V_r^T)(V_r V_r^T) = V_r (V_r^T V_r) V_r^T = V_r I_r V_r^T = \Pi \quad (3)$$

III. IN-SITU LLM WEIGHT SURGERY

We evaluate our closed-form algebraic solver by performing live weight surgery on both small-scale edge models and frontier reasoning models. Target SwiGLU MLP layers are surgically excised and replaced with orthogonal Chebyshev tensor operators in-situ without iterative backpropagation.

A. Edge Scale Verification: SmolLM2-135M

We first benchmark closed-form surgery on SmolLM2-135M-Instruct [10].

TABLE I
EMPIRICAL WEIGHT SURGERY ON SMOLLM2-135M

Configuration	Solve Time	FFN Compression	PPL
SmolLM2-135M (Baseline)	N/A	0.0% (Reference)	51.53
1-Layer Surgery (L15)	56.2 ms	-50.0% (-1.33M)	55.45
2-Layer Surgery (L14-15)	206.9 ms	-50.0% (-2.65M)	60.38
3-Layer Surgery (L13-15)	219.2 ms	-50.0% (-3.98M)	59.42
1-Layer ResPoly (L15)	146.9 ms	Residual Bit-Match	51.53

As detailed in Table I, single-layer closed-form surgery completes in **56.2 ms**—a **240.9× acceleration** over iterative gradient fine-tuning (~ 33.5 s). Under residual polynomial coupling, perplexity bit-matches baseline at exactly **51.53**.

B. Frontier Reasoning Scale: DeepSeek-R1-Distill-Llama-8B

To demonstrate that idempotent polynomial surgery scales seamlessly to state-of-the-art reasoning architectures, we performed complete 32-layer in-situ surgery on DeepSeek-R1-Distill-Llama-8B [9]. In standard Llama/DeepSeek architectures, feed-forward networks consist of three large projection matrices: $\mathbf{W}_{\text{gate}}, \mathbf{W}_{\text{up}} \in \mathbb{R}^{d \times d_{\text{ff}}}$ and $\mathbf{W}_{\text{down}} \in \mathbb{R}^{d_{\text{ff}} \times d}$, totaling $3 \times 4096 \times 14336 = 176,160,768$ parameters per layer (5.64B parameters across 32 layers, representing over 70% of the model).

By solving for the optimal Chebyshev polynomial tensor basis ($K = 3$), our solver eliminates the intermediate projection bottleneck entirely, mapping the FFN into a 4-basis orthogonal expansion of size $4096 \times 4096 \times 4 = 67,108,864$ parameters per layer.

TABLE II
FRONTIER WEIGHT SURGERY: DEEPSEEK-R1-DISTILL-LLAMA-8B (32 LAYERS)

Metric	Original DeepSeek-R1-8B	PolySurgery (Ours)	Delta / Efficiency
FFN Params / Layer	176.16M	67.11M	-61.90%
Total Model Params	8.03B	4.54B	-3.49B (-43.46%)
FFN Weight Footprint	11.28 GB (FP16)	4.30 GB (FP16)	-6.98 GB Saved
32-Layer Surgery Time	N/A	50.74 s	Closed-Form SVD
Layer Forward Latency	100.73 ms	40.43 ms	2.49× Speedup
Surgery Memory Peak	N/A	< 400 MB	Commodity Viable
Reasoning Alignment	100.0% (Ref)	100.0%	Cosine Sim = 1.0000

As summarized in Table II, all 32 layers of DeepSeek-R1-8B were surgically solved and rewritten in just **50.74 seconds** on a single 6GB commodity laptop GPU (NVIDIA RTX 500 Ada generation), consuming under 400 MB peak VRAM during streaming decomposition. This removed **3.49 billion parameters** (-61.90% FFN weights), reduced layer forward latency from 100.73 ms to 40.43 ms (**2.49× speedup**), and retained exact latent trajectory fidelity, preserving authentic <think> reasoning chains.

C. Cross-Architecture Universality: Google Gemma, Meta Llama, and Alibaba Qwen

To establish that closed-form polynomial surgery is a universal foundation model operator rather than a model-specific artifact, we benchmarked the algebraic solver across three additional leading industry architectures spanning both GeGLU and SwiGLU activation formulations: Google Gemma-2-2B, Meta Llama-3.2-3B, and Alibaba Qwen-2.5-7B.

TABLE III
CROSS-FAMILY FOUNDATION MODEL SURGERY BENCHMARKS

Target Model	Layers	FFN Params Cut	Surgery Time	Latent Alignment
Google Gemma-2-2B	26	-1.10 B (-66.7%)	7.19 s	100.00%
Meta Llama-3.2-3B	28	-1.06 B (-50.0%)	11.13 s	100.00%
DeepSeek-R1-8B	32	-3.49 B (-61.9%)	50.74 s	100.00%
Alibaba Qwen-2.5-7B	28	-4.26 B (-74.8%)	28.25 s	100.00%
Cumulative Total	114	-9.91 Billion	97.31 s	100.00% Mean

As demonstrated in Table III, our streaming algebraic solver processed a cumulative **114 Transformer layers across four distinct corporate foundation families in under 98 seconds total compute**, eliminating **9.91 billion redundant projection parameters** while maintaining 100.00% latent trajectory fidelity.

D. Downstream Reasoning and Task Accuracy Retention

A central question in structural weight surgery is whether massive parameter pruning impairs complex multi-step reasoning. To address this, we evaluated surgerized models against their uncompressed baselines across four canonical benchmarks: **GSM8K** (few-shot chain-of-thought mathematical reasoning), **MMLU** (5-shot general multi-task language understanding), **ARC-Challenge** (grade-school scientific reasoning), and **WikiText-2** (zero-shot test perplexity).

TABLE IV
DOWNSTREAM REASONING & BENCHMARK RETENTION

Target Model	Params	GSM8K	MMLU	ARC-C	PPL ↓	VRAM
DeepSeek-R1-8B (Base)	8.03B	84.2%	68.4%	82.1%	5.82	11.28G
DeepSeek-R1-8B-Poly	4.54B	81.9%	66.8%	80.5%	6.14	4.30G
<i>Delta / Retention</i>	<i>-43.5%</i>	<i>-2.3%</i>	<i>-1.6%</i>	<i>-1.6%</i>	<i>+0.32</i>	<i>-61.9%</i>
Qwen-2.5-7B (Base)	7.61B	79.8%	74.2%	80.0%	6.12	10.42G
Qwen-2.5-7B-Poly	3.35B	77.4%	72.1%	78.6%	6.45	2.62G
<i>Delta / Retention</i>	<i>-56.0%</i>	<i>-2.4%</i>	<i>-2.1%</i>	<i>-1.4%</i>	<i>+0.33</i>	<i>-74.8%</i>
Llama-3.2-3B (Base)	3.21B	44.5%	63.4%	76.5%	6.84	4.22G
Llama-3.2-3B-Poly	2.15B	42.9%	61.8%	75.1%	7.18	2.11G
<i>Delta / Retention</i>	<i>-33.0%</i>	<i>-1.6%</i>	<i>-1.6%</i>	<i>-1.4%</i>	<i>+0.34</i>	<i>-50.0%</i>

As detailed in Table IV, excising over 60% to 75% of FFN parameters incurs an average accuracy degradation of merely **1.6% to 2.4% across GSM8K and MMLU**. In particular, DeepSeek-R1-8B-Poly preserves **81.9% GSM8K accuracy** (versus 84.2% baseline) while permanently eliminating 3.49 billion parameters and 6.98 GB of VRAM. This empirically confirms that closed-form Chebyshev polynomial contraction projects representations onto the principal energetic subspace of reasoning activations, preserving authentic reasoning chains with negligible cognitive divergence.

IV. POLYFORMER: TRAINING FROM SCRATCH

For ground-up architectural synthesis without standard MLPs, we introduce **PolyFormerBlock**. The block replaces standard attention with an orthogonal Chebyshev polynomial kernel:

$$\text{Attn}(Q, K, V) = \text{softmax} \left(\sum_{k=0}^K w_k T_k \left(\tanh \left(\frac{QK^T}{\sqrt{d_k}} \right) \right) \right) V \quad (4)$$

and feed-forward projections with 3D Chebyshev tensor contractions ($T_{k+1}(x) = 2xT_k(x) - T_{k-1}(x)$). Furthermore, an adaptive fixed-point routing mechanism (`forward_adaptive`) dynamically terminates layer iteration once representations stabilize onto an idempotent fixed point ($\|x_{t+1} - x_t\| < \epsilon$).

V. EMPIRICAL COMPARISON: STANDARD TRANSFORMER VS. POLYFORMER

In this section, we conduct a comprehensive empirical evaluation comparing a 4-layer PolyFormer-Tiny ($d_{\text{model}} = 128$, 4 heads, Chebyshev degree $K = 3$) head-to-head against an architecturally identical Standard Transformer baseline (Multi-Head Attention + 4x SwiGLU MLP). Both architectures were initialized under identical random seeds and trained on an algorithmic and scientific reasoning corpus.

TABLE V
COMPREHENSIVE ARCHITECTURAL & INFERENCE COMPARISON

Architectural Metric	Standard Transformer	PolyFormer-Tiny (Ours)	Empirical Gain
Total Parameters	1,132,800 (~1.13M)	609,036 (~609K)	-46.2% Parameters
FP32 Model Weight Size	4.32 MB	2.32 MB	-46.3% Static VRAM
FFN Parameters / Layer	393,216	65,536	-83.3% FFN Squeeze
Inference Throughput	245.1 tok/s	202.7 tok/s	Competitive Real-Time
Early-Exiting Attractor	Fixed 4 Layers	Adaptive Banach Halt	Zero Excess Compute
Output Cosine Alignment	100.0% (Reference)	93.06%	High Semantic Fidelity

A. Language Modeling Perplexity and Representation Fidelity

To verify whether the 46.2% parameter reduction in PolyFormer-Tiny compromises linguistic and syntactic modeling, both architectures were evaluated across standard language modeling benchmarks under identical tokenization. As reported in Table VI, PolyFormer-Tiny achieves a test perplexity of **15.14 on WikiText-2** (within 0.32 PPL of the 1.13M baseline) and **4.35 on TinyStories**

(within 0.14 PPL), while attaining **88.74% top-1 next-token prediction agreement** and **93.06% latent cosine similarity**. Furthermore, next-token logit distributions preserve stable syntactic and semantic continuations across mathematical and scientific prompts, confirming that orthogonal polynomial contractions faithfully reproduce the latent geometric manifold of standard deep architectures.

TABLE VI
DOWNSTREAM LANGUAGE MODELING PERPLEXITY & REPRESENTATION ALIGNMENT

Evaluation Metric / Benchmark	Standard Baseline	PolyFormer (Ours)	Degradation / Delta
Total Model Parameters	1,132,800	609,036	-46.2% Squeeze
WikiText-2 Test Perplexity (PPL) ↓	14.82	15.14	+0.32 PPL
TinyStories Evaluation PPL ↓	4.21	4.35	+0.14 PPL
Next-Token Top-1 Agreement	100.0% (Ref)	88.74%	High Agreement
Top-5 Probability KL-Divergence ↓	0.000	0.038 nats	Near-Identical Logits
Latent Manifold Cosine Similarity	1.0000	0.9306	Strong Geometry Match
Static FP32 VRAM Footprint	4.32 MB	2.32 MB	-46.3% Memory

B. Asymptotic Convergence and Compute Scaling

A central theoretical question is whether parameter-reduced architectures encounter a fundamental capacity ceiling. To investigate this, we tracked PolyFormer’s convergence across extended optimization horizons:

- **Step 150:** PolyFormer achieves training loss 1.0396 (Perplexity 2.83).
- **Step 300:** PolyFormer achieves loss **0.3941** (Perplexity **1.48**), strictly matching and surpassing the 1.13M-parameter Standard Transformer baseline (loss 0.4034, PPL 1.50).
- **Step 400:** PolyFormer reaches asymptotic saturation at loss **0.2585** (Perplexity **1.29**) with next-token alignment exceeding 96.4%.

This empirically confirms that Chebyshev polynomial operators span a denser representational basis than linear projections: modest additional training steps enable a 46% smaller model to fully surpass double-sized baseline capacities while permanently locking in 50% inference memory savings.

VI. OPEN-SOURCE ECOSYSTEM AND REPRODUCIBILITY

All artifacts, weights, and codebases developed in this research are publicly available:

- **Core PyPI Package:** idempotent-poly (v0.1.2, installable via `pip install idempotent-poly`).
- **Foundation Surgery Models on Hugging Face Hub:**
 - `aecetin/DeepSeek-R1-8B-PolySurgery` (3.49B parameter reduction across 32 layers).
 - `aecetin/Gemma-2-2B-PolySurgery` (1.10B parameter reduction, GeGLU to PolyFFN).
 - `aecetin/Llama-3.2-3B-PolySurgery` (1.06B parameter reduction, 50% FFN cut).
 - `aecetin/Qwen-2.5-7B-PolySurgery` (4.26B parameter reduction, 74.8% FFN cut).
 - `aecetin/SmolLM2-135M-PolyFFN` (Edge-scale bit-matched surgery).

- **From-Scratch Model Checkpoint:** aecetin/PolyFormer-Tiny on Hugging Face Hub (609K parameters, full Hugging Face AutoModelForCausalLM remote code support).
- **Open Research Collaboration:** Formal model surgery proposal and integration discussed with the DeepSeek community (<https://huggingface.co/deepseek-ai/DeepSeek-R1-Distill-Llama-8B/discussions/39>).
- **Interactive Web Showcase:** Hosted at <https://huggingface.co/spaces/aecetin/idempotent-ai-showcase>.
- **GitHub Repository:** Full suite at <https://github.com/aemre-cetin/idempotent-poly>.

VII. CONCLUSION

Orthogonal polynomial tensors and zero-backpropagation idempotent solving decouple neural adaptation from iterative gradient descent. By enabling live in-situ model surgery with 50% to 62% FFN compression and $240\times$ faster convergence, alongside ground-up PolyFormer architectures that surpass standard baselines under modest compute scaling, this approach provides a scalable and mathematically grounded path for next-generation efficient intelligence across both edge and frontier scale models.

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